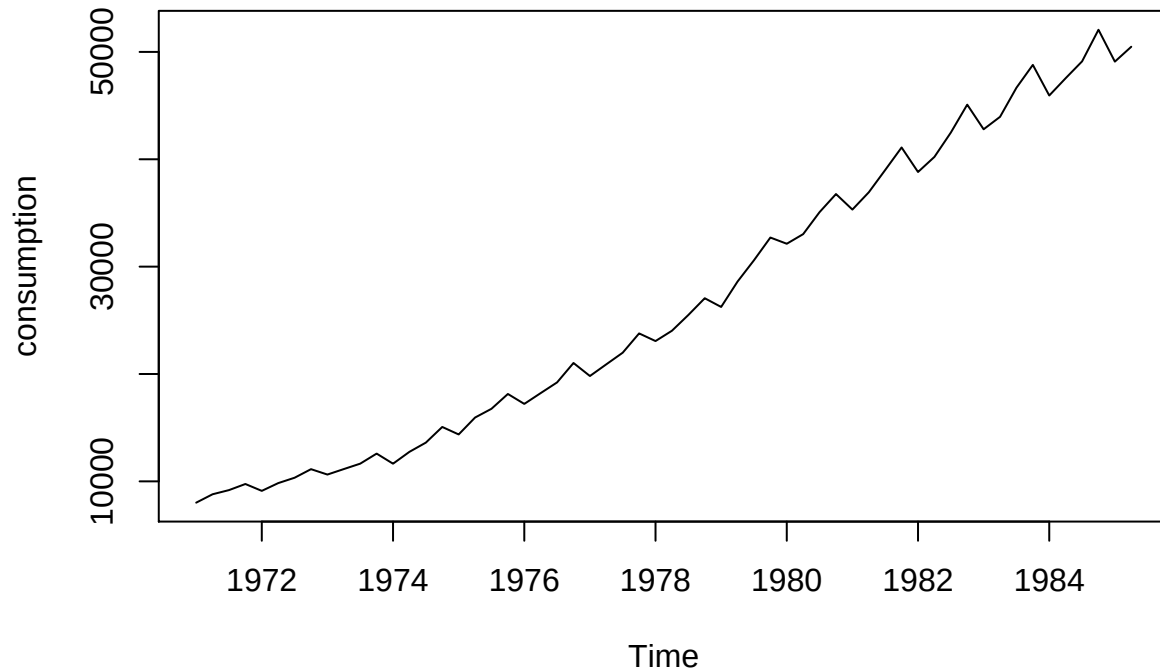


HW3 – Ying Zheng

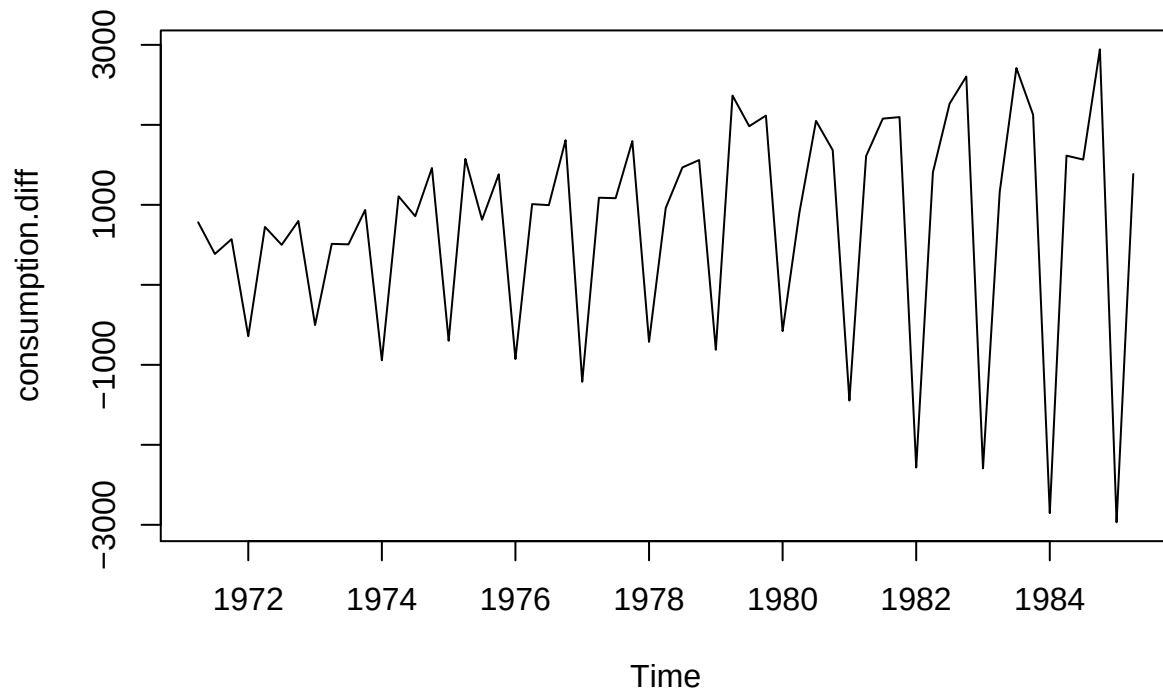
Part I - Problem 1

```
data(IncomeUK)
consumption = IncomeUK[,2]
plot(consumption)
```



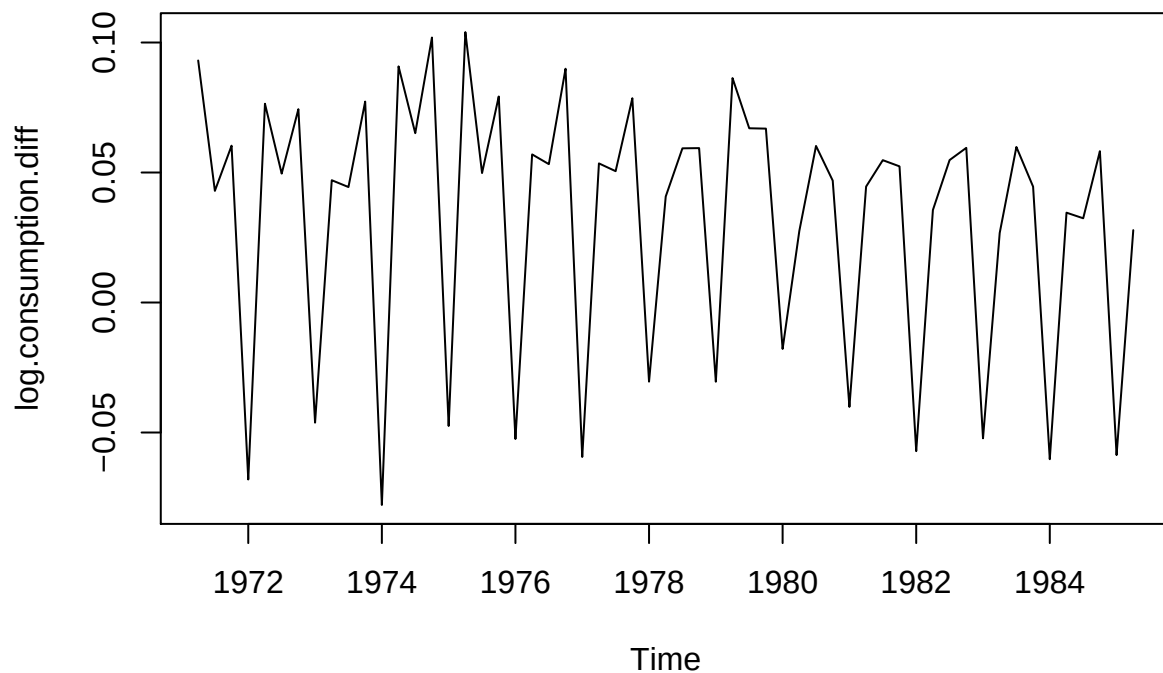
Clearly, there is an upward going trend in the consumption time series with a seasonality of approximately one year. Within each cycle, consumption first keeps moving up for around 3/4 of the time and goes down in the end. To remove the trend, I use the first-differenced series, and inspect the plot of the series, along with the result of the ADF test.

```
consumption.diff = diff(consumption,1, 1)
plot(consumption.diff)
```



We are able to remove the trend by taking first order difference but it still seems not stationary (variance getting larger as time getting larger). So we try to take difference of log consumption.

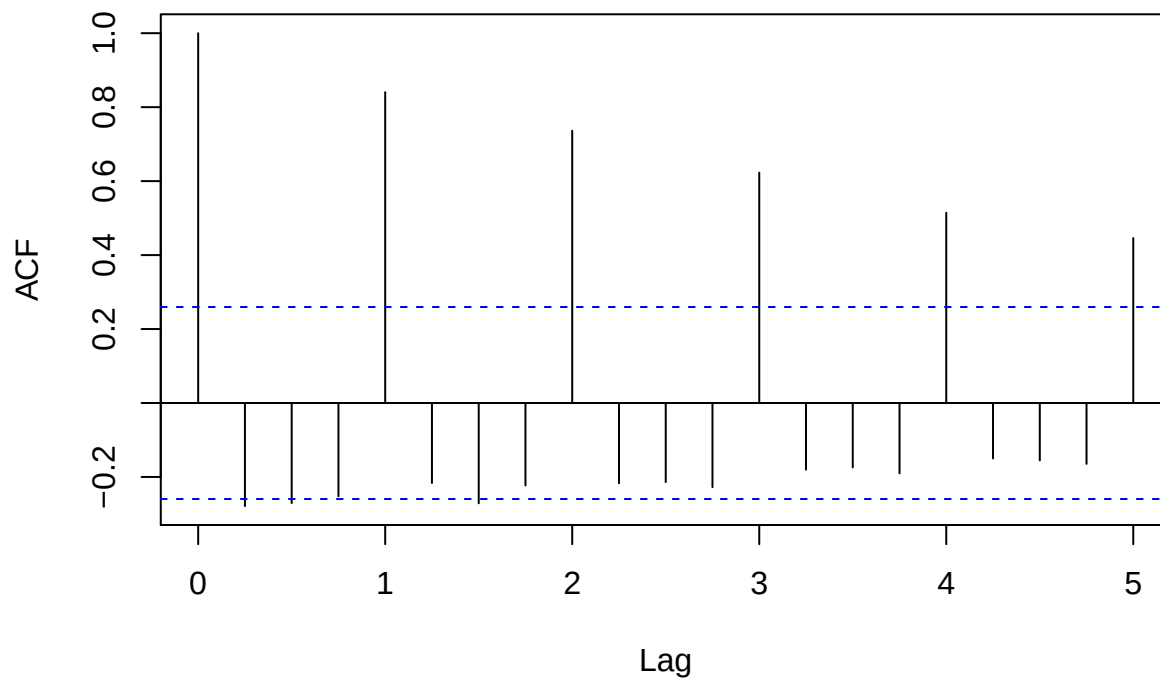
```
log.consumption.diff = diff(log(consumption),1, 1)
plot(log.consumption.diff)
```



Now the time series looks stationary. Next, we do ACF on both consumption difference and log consumption difference:

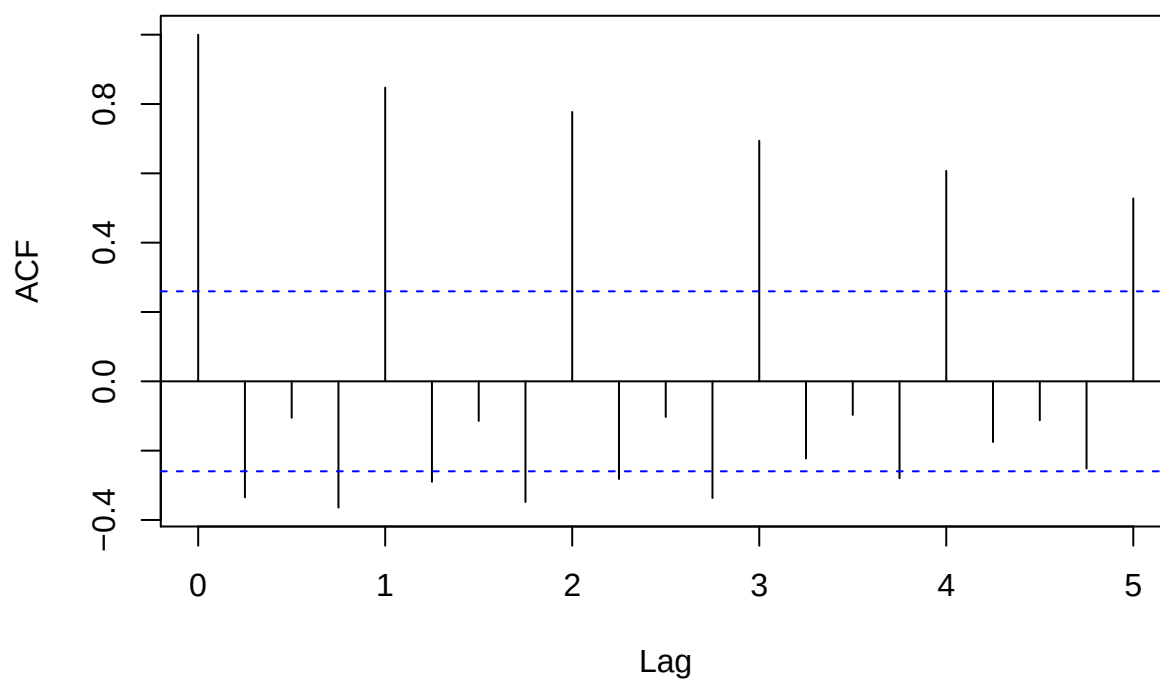
```
acf(consumption.diff, lag = 20)
```

Series consumption.diff



```
acf(log.consumption.diff, lag = 20)
```

Series log.consumption.diff



Both the row data and ACF consumption differencing suggest a clear seasonal pattern. So we should do seasonal differencing. Also, it's easy to see that log transformation gives a more stationary time series. So we should fit seasonal ARIMA model to the data with a log transformation.

Problem 2

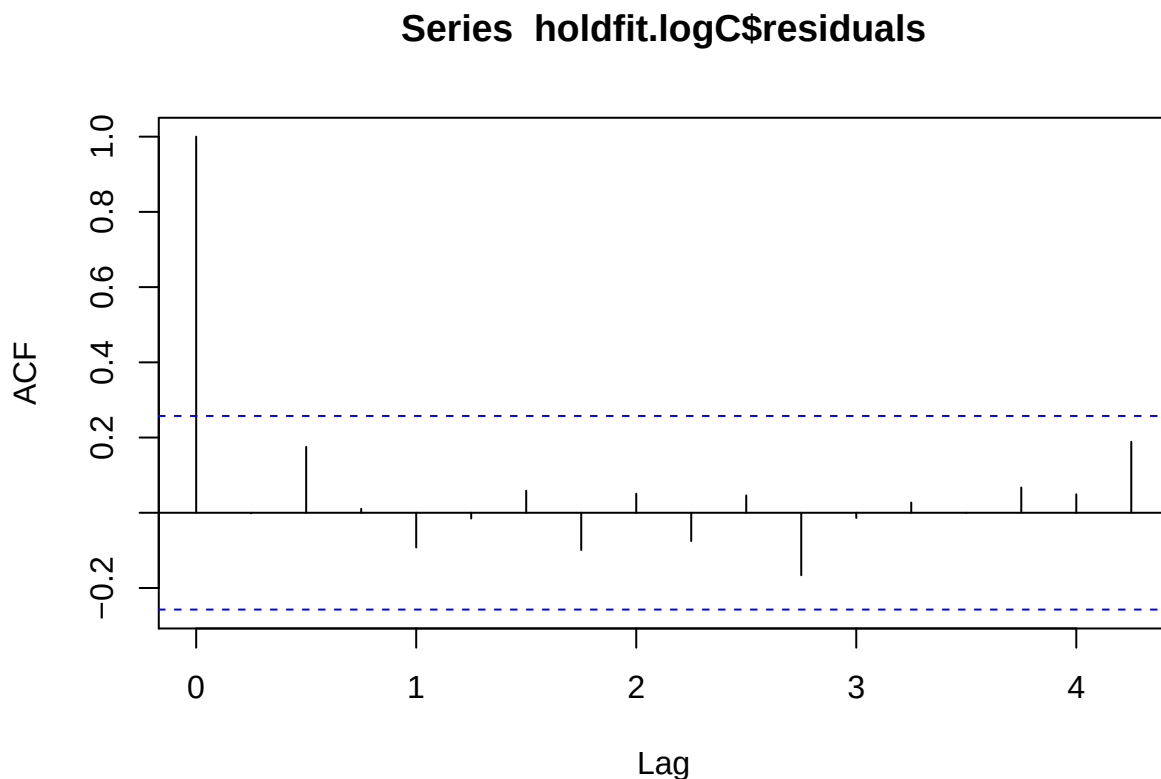
```
# By observing the time series, it's easy to find period = 4
log.consumption = ts(log(consumption), frequency = 4)
holdfit.logC = auto.arima(log.consumption, approx = F, stepwise = F)
holdfit.logC
```

```
## Series: log.consumption
## ARIMA(0,1,0)(0,1,1)[4]
##
## Coefficients:
##          sma1
##        -0.5348
## s.e.    0.1164
##
## sigma^2 estimated as 0.0003056: log likelihood=139.77
## AIC=-275.55   AICc=-275.31   BIC=-271.61
```

By looking at the result, we can tell that: for the non-seasonal part, the order of autoregression is 0, the degree of differencing is 1 and the order of moving average is 0; for the seasonal part (with period of 4), the order of autoregression is 0, the degree of differencing is 1 and the order of moving average is 1.

Problem 3

```
model.acf=acf(holdfit.logC$residuals)
```



There is no significant statistical evidence to reject that the residuals have no correlations.

Part II

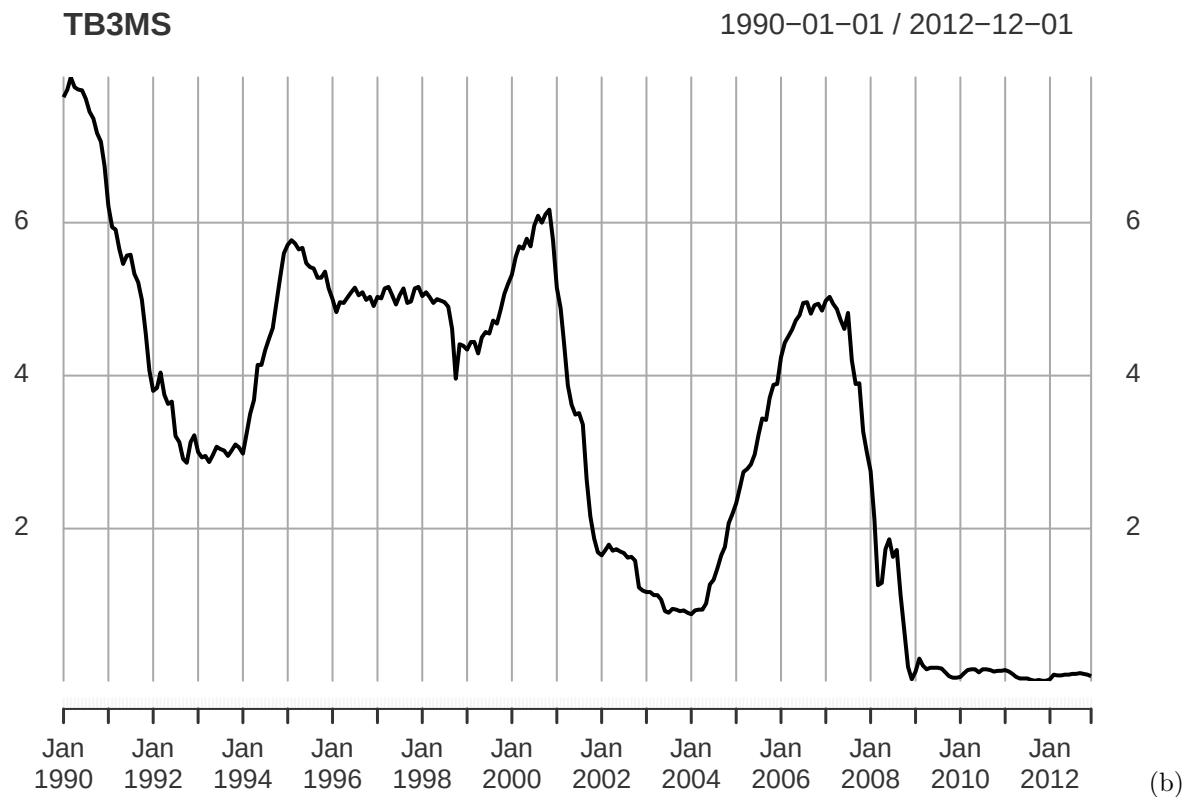
```
TB3MS = getSymbols("TB3MS",src = "FRED",auto.assign = F)
```

```
## 'getSymbols' currently uses auto.assign=TRUE by default, but will
## use auto.assign=FALSE in 0.5-0. You will still be able to use
## 'loadSymbols' to automatically load data. getOption("getSymbols.env")
## and getOption("getSymbols.auto.assign") will still be checked for
## alternate defaults.
##
## This message is shown once per session and may be disabled by setting
## options("getSymbols.warning4.0"=FALSE). See ?getSymbols for details.
```

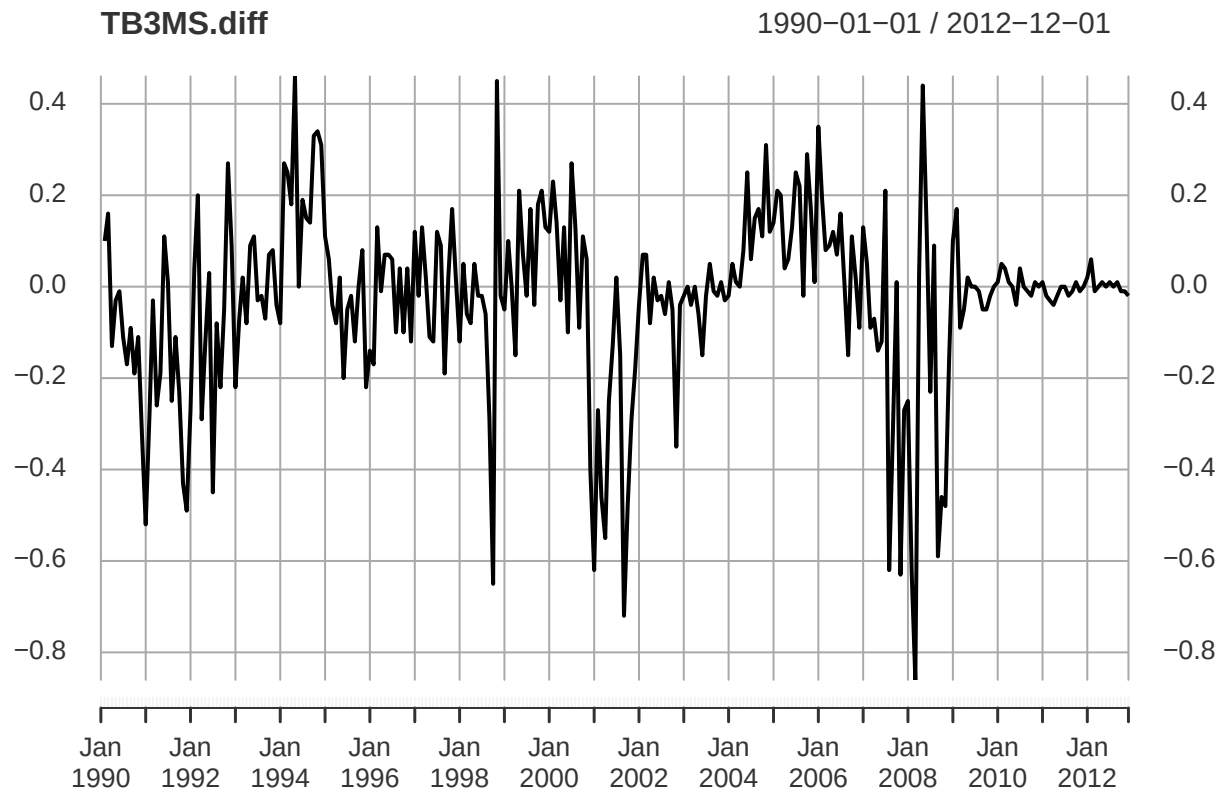
```
TB3MS = TB3MS["1990::2012"]
```

(a) The plot does not seem stationary.

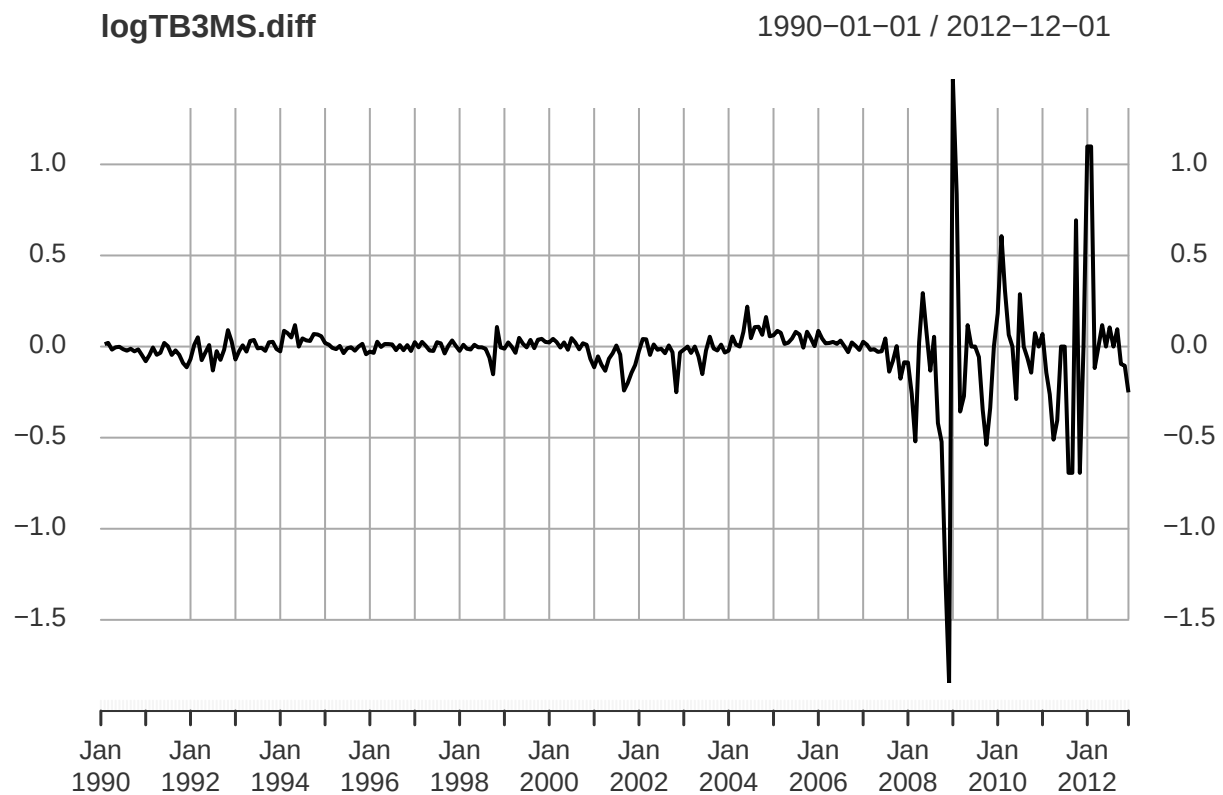
```
plot(TB3MS)
```



```
TB3MS.diff = diff(TB3MS, 1, 1)
plot(TB3MS.diff)
```



```
logTB3MS.diff = diff(log(TB3MS), 1, 1)
plot(logTB3MS.diff)
```



```
adf.test(as.numeric(TB3MS.diff[-1]), alternative = "s")
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: as.numeric(TB3MS.diff[-1])  
## Dickey-Fuller = -3.5576, Lag order = 6, p-value = 0.03763  
## alternative hypothesis: stationary
```

By looking at first order difference on both row and transformed data, it seems that the not transformed one is more stationary. We did ADF test on that and got a p value of 0.03763, which is less than 0.05. So we accept the alternative hypothesis that the difference on row data is stationary.

(c) The order of autoregression is 3, and the order of moving average is 1.

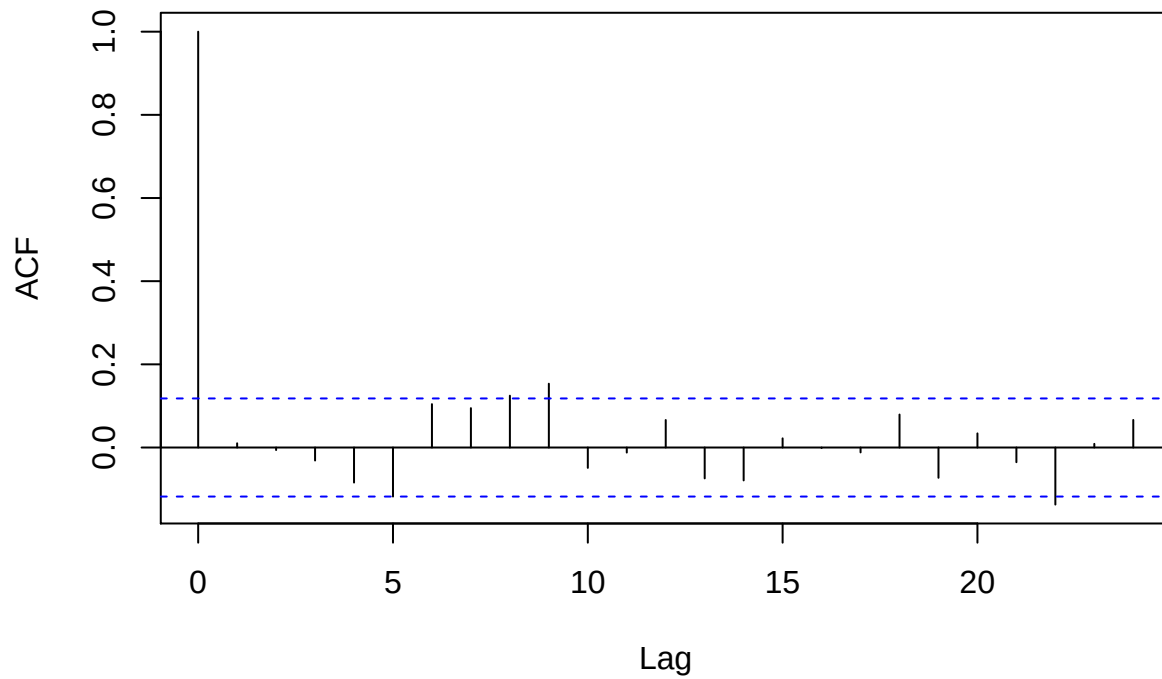
```
holdfit.TB3MS = auto.arima(TB3MS, approximation = F, stepwise = F)  
holdfit.TB3MS
```

```
## Series: TB3MS  
## ARIMA(3,1,0)  
##  
## Coefficients:  
##          ar1          ar2          ar3  
##          0.4093   -0.0300   0.2523  
## s.e.    0.0582    0.0633   0.0582  
##  
## sigma^2 estimated as 0.02894: log likelihood=98.17  
## AIC=-188.34   AICc=-188.19   BIC=-173.87
```

(d) We tried acf plot, box test and qq plot. ACF plot suggests some correlation in further lags, box test gives a p value less than 0.05 and qq plot shows a trend deviates from the 45 degree line. Since none of the diagnostic tests on the residuals passed, the model did not seem to be a reasonable fit.

```
acf(holdfit.TB3MS$residuals)
```

Series holdfit.TB3MS\$residuals

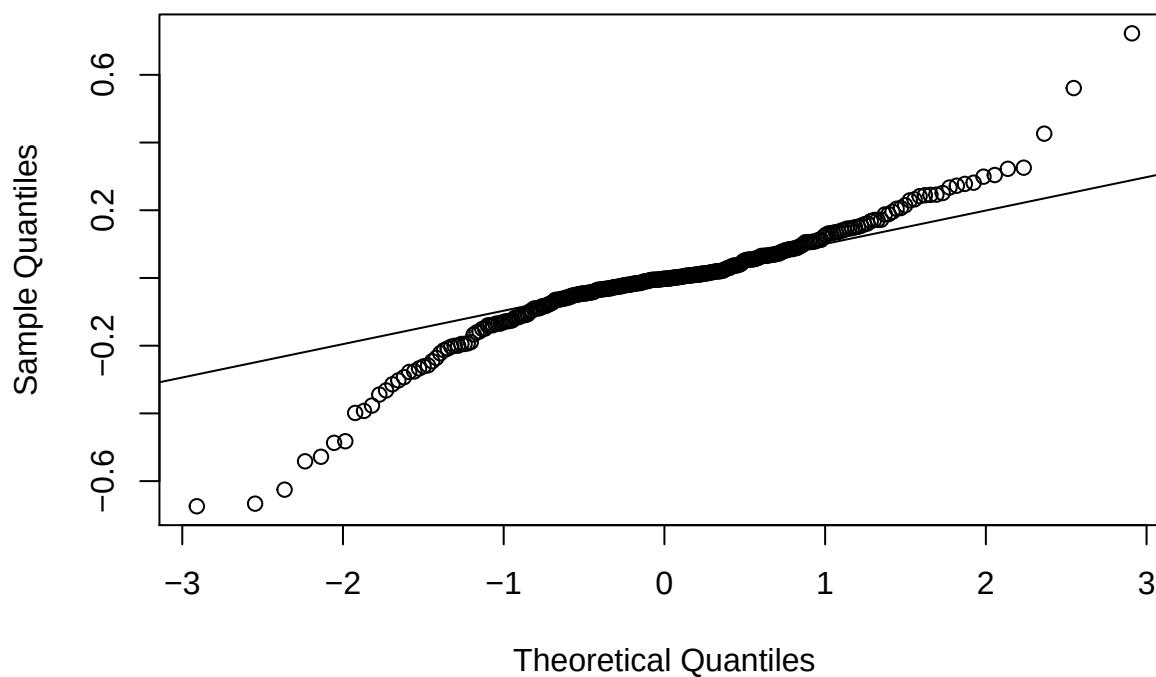


```
Box.test(holdfit.TB3MS$residuals, lag = 20, type="Ljung-Box", fitdf = 3)
```

```
##  
## Box-Ljung test  
##  
## data: holdfit.TB3MS$residuals  
## X-squared = 32.57, df = 17, p-value = 0.01277
```

```
qqnorm(holdfit.TB3MS$residuals)  
qqline(holdfit.TB3MS$residuals)
```

Normal Q-Q Plot



(e)

```
holdpreds = predict(holdfit.TB3MS, n.ahead = 3)
CI.1 = c(holdpreds$pred[1]-1.96*holdpreds$se[1],holdpreds$pred[1]+1.96*holdpreds$se[1])
CI.2 = c(holdpreds$pred[2]-1.96*holdpreds$se[2],holdpreds$pred[2]+1.96*holdpreds$se[2])
CI.3 = c(holdpreds$pred[3]-1.96*holdpreds$se[3],holdpreds$pred[3]+1.96*holdpreds$se[3])
CI.1
```

```
## [1] -0.2738443  0.3930244
```

```
CI.2
```

```
## [1] -0.5227881  0.6295988
```

```
CI.3
```

```
## [1] -0.7271855  0.8194638
```

(f) Yes, the three intervals include the observed values.

```
TB3MS.all = getSymbols("TB3MS",src = "FRED",auto.assign = F)
TB3MS.all["2013"][1:3]
```

```
##          TB3MS
## 2013-01-01  0.07
## 2013-02-01  0.10
## 2013-03-01  0.09
```